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DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Urban Green Equity in the 15-Minute City: A Data-Driven Framework for Spatial Justice in Milan

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Abstract

Urban green space is essential for public health, environmental resilience, and social cohesion, yet access is often uneven, raising concerns of spatial justice. Many planning metrics emphasize aggregate provision, overlooking walkability, population demand, and intra-urban disparities. This study presents a spatially explicit, data-driven framework to assess walkable green equity in Milan. The model integrates a regular spatial grid, network-based pedestrian isochrones, and census-derived population data to compute the Green Equity Ratio (GER), a composite indicator of the balance between green accessibility and demographic intensity. Results, aggregated at the neighborhood (*Nucleo di Identità Locale*, NIL) scale and analyzed through clustering, reveal marked disparities: central and semicentral NILs combine high population pressure with low access, while many peripheral areas show favorable conditions. The resulting typology supports targeted, need-based policy interventions. The approach is transferable to other cities and public services, providing a replicable tool for evidence-based, equity-oriented urban planning.

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1 Introduction

Urban areas worldwide face increasing pressures from spatial inequality, environmental vulnerability, and declining quality of life. Among the public goods at stake, urban green space functions not only as a recreational amenity but also as critical infrastructure that supports public health, social cohesion, and climate resilience [1, 2]. Empirical evidence links access to urban greenery with benefits such as improved mental well-being, mitigation of urban heat, and promotion of social interaction [3]. Yet these benefits are unevenly distributed: disadvantaged populations often experience lower access to high-quality green environments, reinforcing broader patterns of socio-spatial inequality [4]. This uneven geography of access raises concerns about spatial justice, understood as the fair allocation of resources and opportunities across the urban landscape [5].

In this context, the ability to reach green spaces on foot is a critical dimension of everyday livability and equity. Walkable access enhances inclusivity, particularly for those with limited mobility or without private transport, and aligns with the functional logic of the “15-Minute City” paradigm [6]. However, many cities still lack analytical tools that assess such access in a spatially explicit and demographically sensitive manner. Traditional planning metrics—such as green square meters per capita—offer useful benchmarks but fail to capture the relational nature of spatial equity. Green infrastructure may meet aggregate targets yet remain inaccessible to significant portions of the population due to its location, connectivity, or design [7]. Common analytical approaches often rely on static buffers or Euclidean distances applied to aggregated units, overlooking the actual structure of pedestrian networks and the spatial heterogeneity of population distribution. Furthermore,

socio-demographic variation is frequently ignored, treating all residents as equally positioned with respect to access.

Addressing these shortcomings requires models that integrate urban form, infrastructure connectivity, and demographic dynamics into a coherent framework for evaluating territorial justice. Such models should incorporate: (i) the real configuration of walkable networks; (ii) spatial variation in residential population; and (iii) proximity-based thresholds that reflect the temporal scale of daily life.

In response to these gaps, this study develops a spatially explicit, data-driven framework for evaluating walkable green equity in Milan. The model combines the pedestrian network with the distribution of public green areas and residential population to assess whether infrastructure supply is proportionally available relative to local demographic demand. Grounded in the principles of spatial justice and the 15-Minute City, the analysis measures access within a 15-minute walking threshold.

A composite indicator—the Green Equity Ratio (GER)—is introduced to relate green accessibility (GA) to population intensity (PI) at the neighborhood (Nucleo di Identità Locale, NIL) scale. This diagnostic metric identifies areas where provision fails to match demand, highlighting both deficits and surpluses. Unsupervised clustering of NILs based on GA and PI produces typologies that capture recurring accessibility–demand configurations, enabling pattern-based interpretation and supporting differentiated policy responses.

The remainder of the paper is structured as follows: Section 2 presents the conceptual foundations and relevant literature; Section 3 describes data sources, indicators, and analytical workflow; Section 4 reports empirical findings, including indicator distributions and NIL typologies; and Section 5 discusses the policy implications, methodological limitations, and avenues for more inclusive, responsive urban planning.

2 Background and Related Work

This section provides the conceptual and analytical context for the study. It begins by discussing the role of urban green equity as a key dimension of spatial justice, focusing on the importance of fair access to green infrastructure in dense urban environments. It then introduces the 15-Minute City as a spatial model that supports proximity-based planning and highlights its relevance for addressing local disparities in access to public goods. The final part reviews recent literature on green accessibility and spatial equity, identifying key contributions as well as current methodological limitations.

2.1 Urban Green Equity and Spatial Justice

Urban green space is increasingly recognized as a critical component of public infrastructure in contemporary cities. Beyond its aesthetic function, it delivers essential ecosystem services and supports multiple dimensions of urban life. Well-designed and accessible green areas contribute to physical and mental health, mitigate urban heat island effects, absorb pollutants, and provide space for recreation and social interaction [1, 8, 9]. These functions position green space alongside mobility, water, and energy systems as part of the functional core of the urban environment. In dense metropolitan contexts, where access to private open space is limited, public green becomes a key determinant of everyday livability and climate resilience; its spatial distribution shapes patterns of access, exposure, and opportunity across the urban population [10, 11, 12].

Equity in urban contexts refers to the fair allocation of resources, opportunities, and services based on differentiated local needs. Unlike equality, which implies uniform distribution, equity accounts for social, spatial, and economic asymmetries across the population [13, 1]. In practice, this means recognizing that not all areas or groups require the same amount or type of provision to achieve comparable levels of well-being.

Spatial justice provides a normative framework for evaluating how urban resources are distributed across space and how this distribution affects different social groups. The concept expands the idea of equity by embedding it within the geography of the city, focusing on who has access to what, where, and under which conditions [14]. In urban studies, spatial justice is often articulated through three dimensions: distribution, accessibility, and participation [15, 16]. Together, these dimensions provide a critical lens for assessing whether cities enable or constrain equitable access to public goods such as green space.

Translating green equity into urban policy requires more than normative commitment—it demands operational metrics. Without measurement, the concept remains abstract and difficult to act upon. Identifying spatial disparities in green provision involves quantifying both the availability of accessible green space and the demographic demand it is meant to serve [17, 18].

Assessing equity in this context requires a relational perspective: green access must be evaluated in proportion to local population distribution and intensity. This shift from absolute coverage to contextual adequacy enables a more precise understanding of who benefits from public green infrastructure and who does not, and provides the analytical foundation for targeting interventions and monitoring their impact over time [17, 19].

2.2 The 15-Minute City as a Spatial Framework

The concept of the 15-Minute City, introduced by Carlos Moreno, proposes a reconfiguration of urban space around the principle of temporal proximity. It envisions a city where all essential functions of daily life—such as housing, work, education, healthcare, commerce, and leisure—can be reached within a 15-minute walk or bike ride from any residence [6]. The model promotes a polycentric and human-scaled urban form, in contrast to car-dependent and functionally segregated layouts; core principles include proximity, land-use diversity, social inclusion, sustainability, and active mobility [6, 20]. By reorganizing existing assets rather than adding new infrastructure, the approach aims to enhance quality of life, reduce transport emissions, and foster local social interaction [21, 22].

The proximity principle embedded in the 15-Minute City aligns closely with equity goals: accessibility can be operationalized as the ease with which people reach opportunities, making spatial proximity a measurable proxy for access [23, 24]. Focusing on everyday reachability highlights distributive fairness: the ability to access green space, healthcare, education, and other services within walking distance is uneven across territories, and proximity-based planning makes such disparities visible and actionable [24, 25].

Milan offers a particularly suitable context for applying these principles. Its compact urban form, relatively high residential density, and extensive public transport network support walkability and short-range accessibility; at the same time, the city exhibits heterogeneous distributions of green spaces and services, with concentrations in central districts and more fragmentation at the periphery. Institutionally, Milan is subdivided into *Nuclei di Identità Locale* (NILs), widely used in planning and analytics, which provide a meaningful neighborhood-scale framework for assessing equity in access to urban resources [26, 27]. The combination of a complex spatial structure and an established system of local governance makes Milan a relevant case for evaluating proximity-based strategies through a lens of spatial justice.

2.3 Related Work

Green accessibility has been widely studied in urban planning and public health, often with the aim of quantifying residents’ exposure to green infrastructure [2, 8]. Traditional approaches typically rely on static measures such as circular buffers or straight-line (Euclidean) distances [19, 28]. While these methods provide a general sense of proximity, they overlook the complexity of urban form and pedestrian mobility, which can bias results when street-network structure and route circuitry matter [29, 30]. More advanced models therefore incorporate network-based distances to represent walkable access more realistically, although adoption has historically been constrained by data availability and computational costs [31, 29].

The field has shifted from general provision toward equity-oriented analysis, examining how access intersects with social and demographic variables [7, 32]. Measures increasingly incorporate population density, age, income, or vulnerability to detect mismatches between availability and local demand [33, 34]. Equal distribution does not necessarily yield equitable outcomes, which calls for evaluating access relative to differentiated needs and pre-existing inequalities [17, 2].

Despite progress, key methodological limitations persist. First, reliance on administrative units can mask intra-urban variation due to the MAUP and related context problems [35, 36]. Second, simplified proximity metrics (e.g., Euclidean buffers) can misrepresent walkable reachability compared to network-based measures [29, 37]. Third, aggregation to broad units dilutes fine-scale heterogeneity and may hide high-need areas [17, 28]. Finally, although supply–demand formulations are growing, spatially explicit demand-sensitive approaches and capacity-weighted metrics remain uneven across cases [38, 39].

This study’s contribution. In light of these gaps, this study develops an operational, equity-oriented workflow that: (i) avoids administrative-unit distortions by working on a regular 250 m grid [35, 36]; (ii) measures *walkable* green access via network-based isochrones constructed on the actual pedestrian graph [31, 30]; and (iii) relates access to local demand through a *Green Equity Ratio* (GER), a demand-sensitive indicator informed by pressure/exposure approaches in the equity literature [18, 40]. We then aggregate to NILs and derive interpretable typologies via clustering to support differentiated policy responses. This integrated design advances beyond supply-only or Euclidean-proximity assessments by balancing network-based accessibility with population intensity at fine spatial resolution.

3 Data and Methodology

This section outlines the datasets, spatial preprocessing steps, and analytical methods used to assess walkable green equity in Milan. The workflow integrates multiple open spatial datasets, harmonized to a common spatial

reference system, and employs a network-based approach to model pedestrian accessibility. Public green areas, population data, and administrative boundaries are combined within a regular analysis grid to ensure spatial consistency. Pedestrian isochrones are computed to operationalize functional proximity, forming the basis for three core indicators: *Green Accessibility*, *Population Intensity*, and the *Green Equity Ratio*. Validity rules are applied to filter marginal cells, and results are aggregated to the *Nuclei di Identità Locale* (NIL) scale to align with planning-relevant units. Finally, a clustering procedure groups NILs into typologies based on supply–demand profiles, supporting a comparative interpretation of spatial equity conditions across the city.

3.1 Data Sources

The analysis integrates multiple open spatial datasets describing Milan’s public green areas, administrative boundaries, pedestrian network, and population distribution. All layers were harmonized to a common coordinate reference system (EPSG:4326) to ensure interoperability and avoid misalignments in subsequent processing. Each dataset plays a distinct role in the analytical framework: green areas define the spatial supply, population data capture demographic demand, NIL boundaries support neighborhood-scale aggregation, and the pedestrian network enables network-based walkability modeling.

3.1.1 Parks

Public green areas were extracted from OpenStreetMap [41] using the tags `leisure=park` and `leisure=garden`. Only polygon geometries were retained, ensuring compatibility with area-based accessibility calculations. Features smaller than 5,000 m² were excluded, following common practice in accessibility studies that focus on functionally relevant parks rather than marginal or decorative patches [34, 40]. This filtering step improves interpretability by removing residual elements such as narrow strips or incidental greenery that are unlikely to provide meaningful public use.

3.1.2 City Boundary and Grid

The spatial extent of the study corresponds to the official administrative boundary of the Municipality of Milan. To provide a uniform analytical framework, the urban area was partitioned into a regular grid of 250 m × 250 m cells. This resolution offers a balance between spatial detail and computational feasibility, and has been adopted in previous urban accessibility assessments [25]. The use of a regular tessellation avoids the shape biases of irregular administrative units and enables scalable aggregation from fine-grained measures to higher-level zones.

3.1.3 Census Sections and Population

Population data were sourced from the 2021 National Census provided by ISTAT, supplied as polygon shapefiles representing census sections (*sezioni di censimento*) with an attribute field (POP21) reporting resident counts. Census sections are the smallest official statistical units in Italy and, in dense urban contexts, allow for a detailed representation of residential patterns [42]. These data underpin the demand-side analysis by enabling localized comparisons between population distribution and walkable green supply, consistent with the equity-oriented perspective of the 15-minute city.

3.1.4 NIL

Boundaries for Milan’s *Nuclei di Identità Locale* (NILs) were obtained from the Municipality’s open data portal, as defined in the current Urban Planning Document (PGT 2030). NILs are sub-municipal units designed to capture socio-territorial cohesion and local identity. They are widely used in Milan’s planning practice for spatial diagnostics and resource allocation. In this study, NILs serve as the aggregation level for grid-based indicators, providing an intermediate spatial scale that combines policy relevance with finer-grained analytical inputs.

3.1.5 Pedestrian Network

The pedestrian network was extracted from OpenStreetMap using the `osmnx` Python library [30], with the parameter `network_type='walk'`. The dataset includes all walkable links within the municipal boundary, represented as a directed graph of nodes and edges. This network model captures the actual configuration of pedestrian infrastructure—street connectivity, crossings, and walkable paths—and forms the basis for computing network-based isochrones. Such an approach has become standard in accessibility research, as it reflects real travel conditions more accurately than Euclidean buffers [29, 31].

3.2 Pedestrian Isochrones

An isochrone delineates the area reachable from a given point within a specified time or distance, representing a continuous surface of accessibility shaped by the underlying transport network [23, 24]. Unlike simple Euclidean buffers, pedestrian isochrones capture the actual walkable environment by accounting for street geometry, connectivity, and barriers [30]. This network-based representation enables a realistic estimation of reachable areas, which is critical for measuring local access to urban green space within the framework of the 15-minute city.

In this study, each isochrone originates from the centroid of a 250 m grid cell, projected onto the pedestrian network by snapping to the nearest walkable node. The network was derived from OpenStreetMap using `osmnx` [30], ensuring alignment between the cell-based spatial model and the street-level topology. A maximum walking distance of 1,250 m—corresponding to a 15-minute walk at an average speed of 5 km/h—was adopted as the functional proximity threshold, in line with proximity-based planning principles [6, 20]. This uniform threshold allows for direct comparison of accessibility levels across the urban grid.

For each origin node, a Dijkstra shortest-path algorithm was applied using edge length as the cost function to identify all reachable nodes within the distance limit. The coordinates of these nodes were then used to generate a convex hull polygon approximating the walkable area. Cells for which no valid isochrone could be computed—either due to inability to project the centroid onto the network or because the origin node fell outside the cell—were excluded from further analysis to preserve the spatial coherence and comparability of accessibility indicators.

3.3 Measures of Green Accessibility and Demand

Three indicators were computed for each 250 m grid cell: *green accessibility*, *population intensity*, and their ratio, the *Green Equity Ratio* (GER). Together, they capture the spatial supply of walkable public green space, the demographic demand, and their balance. These measures form the basis for subsequent aggregation and clustering, enabling a spatially explicit assessment of equity in green provision.

3.3.1 Green Accessibility

In line with proximity-based planning and spatial justice principles [6, 24], the Green Accessibility Index (GA) quantifies the surface of public green space reachable within a 15-minute walk from each grid cell. Eligible green areas were extracted from OpenStreetMap [41], retaining only polygons tagged as `leisure=park` or `leisure=garden` and larger than 5,000 m², following thresholds commonly applied in accessibility studies to exclude marginal or non-functional fragments [34, 40].

For each cell, the valid pedestrian isochrone was intersected with eligible park polygons, and the total accessible area computed. To mitigate the influence of disproportionately large parks, a square root transformation was applied, reflecting diminishing marginal benefits of additional green surface once basic needs are met [19]. The resulting GA serves as the reference measure of walkable green availability for equity analysis.

3.3.2 Population Intensity

Population Intensity (PI) represents the estimated residential population associated with each 250 m grid cell, derived from 2021 ISTAT census data [42]. As census sections do not align with the analysis grid, values were obtained via areal interpolation: each section’s population was proportionally distributed across overlapping cells based on intersection area, preserving census totals.

To mitigate the influence of extremely high population concentrations, a square root transformation was applied to cell-level population counts, reflecting diminishing marginal demand once basic thresholds are met.

PI provides the demand-side counterpart to GA, highlighting areas where concentrated residential presence may indicate higher competition for green resources.

3.3.3 Green Equity Ratio

The Green Equity Ratio (GER) expresses the balance between green accessibility and local population pressure, defined at the grid-cell level as the ratio between *Green Accessibility* (GA) and *Population Intensity* (PI), as introduced in Sections 3.3.1 and 3.3.2.

For each 250 m cell, GER is computed as:

$$GER_{cell} = \frac{GA_{cell}}{PI_{cell}}$$

GER values are then averaged over valid cells within each *Nucleo di Identità Locale* (NIL) to produce a neighborhood-scale diagnostic. High GER values indicate provision exceeding local demand, while low values highlight areas where green availability is likely insufficient relative to resident population.

3.4 Spatial Validity and Aggregation

The indicators in Section 3.3 are computed on a 250 m grid. Not all cells, however, yield meaningful information. We therefore define validity rules at cell level and then aggregate valid observations to NILs to obtain interpretable neighborhood-scale patterns.

3.4.1 Cell Inclusion Criteria

Cells with very low or zero population distort accessibility–demand ratios and add noise. To ensure that indicators represent inhabited urban fabric and that the Green Equity Ratio is numerically stable, we exclude cells with fewer than 25 residents per 250 m cell. This operational threshold corresponds to roughly 400 inhabitants/km², marking the transition from marginal to demographically relevant areas. The filter keeps the analysis focused on locations where actual demand for walkable green space exists.

3.4.2 NIL Aggregation and Filtering

Grid-level indicators are aggregated to *Nuclei di Identità Locale* (NILs) to align with a spatial unit that is both analytically manageable and operationally meaningful for planning. For each NIL, we compute the arithmetic mean over valid cells. To ensure reliability, a NIL is retained only if two conditions hold simultaneously: (i) at least 40% of its grid cells are valid; and (ii) the absolute count of valid cells is ≥ 10 . The percentage threshold guarantees adequate spatial coverage; the absolute threshold prevents averages driven by too few observations.

3.5 Clustering of NIL Typologies

The clustering procedure aims to construct synthetic typologies of NILs based on green accessibility and demographic demand. By grouping neighborhoods with similar profiles, the analysis enables a comparative reading of spatial equity conditions across Milan and supports the design of targeted policy interventions. In the framework of the 15-Minute City, such typologies help identify structurally comparable zones that may benefit from shared planning strategies or require differentiated corrective measures [43, 44].

Two standardized input variables are used: Green Accessibility and Population Intensity. These represent the core dimensions of supply and demand, treated independently to avoid redundancy [44]. The Green Equity Ratio is excluded as it is derived from the other two. All variables are z-score standardized to ensure comparability and equal weighting. Clustering is performed using the K-Means algorithm [45] with Euclidean distance, selected for its efficiency and interpretability in low-dimensional, normalized space.

The optimal number of clusters was evaluated through the elbow method and average silhouette score [46]. Although the three-cluster solution achieved the highest silhouette value, $k = 4$ was selected to improve typological differentiation. The additional cluster captures NILs with intermediate profiles—neither underserved nor privileged—which are relevant from a governance perspective. This choice balances statistical compactness with policy-oriented interpretability.

To account for uncertainty in assignment, silhouette values were examined at the NIL level. Units with a silhouette score below 0.20 were flagged as ambiguous and assigned a secondary label based on proximity to the next closest centroid. This step provides interpretive flexibility, particularly for neighborhoods at the boundary between typological categories.

4 Results

This section presents the main analytical outcomes of the framework. The first part provides a descriptive analysis of the three core indicators—Green Accessibility, Population Intensity, and the Green Equity Ratio—covering their distributions and spatial patterns across Milan. It also examines their interrelationships to highlight structural imbalances between supply and demand. The second part focuses on the clustering results, introducing a typology of NILs based on combined accessibility and demographic conditions. These results form the empirical basis for the policy directions discussed in Section 5.

4.1 Descriptive Analysis of Core Measures

This section examines the three core indicators—Green Accessibility (GA), Population Intensity (PI), and the Green Equity Ratio (GER)—focusing on their distribution, spatial patterns, and interrelations at the NIL level. The analysis first considers each measure independently, then explores their combined behaviour to highlight mismatches between green resource availability and demographic demand.

4.1.1 Green Accessibility: distribution and spatial patterns

Green Accessibility shows marked variability across Milan’s NILs (Figure 1). Most neighborhoods fall within intermediate values, with a small subset displaying particularly high or low accessibility. The right-skewed distribution indicates that only a limited number of NILs enjoy consistently high access to public green space.

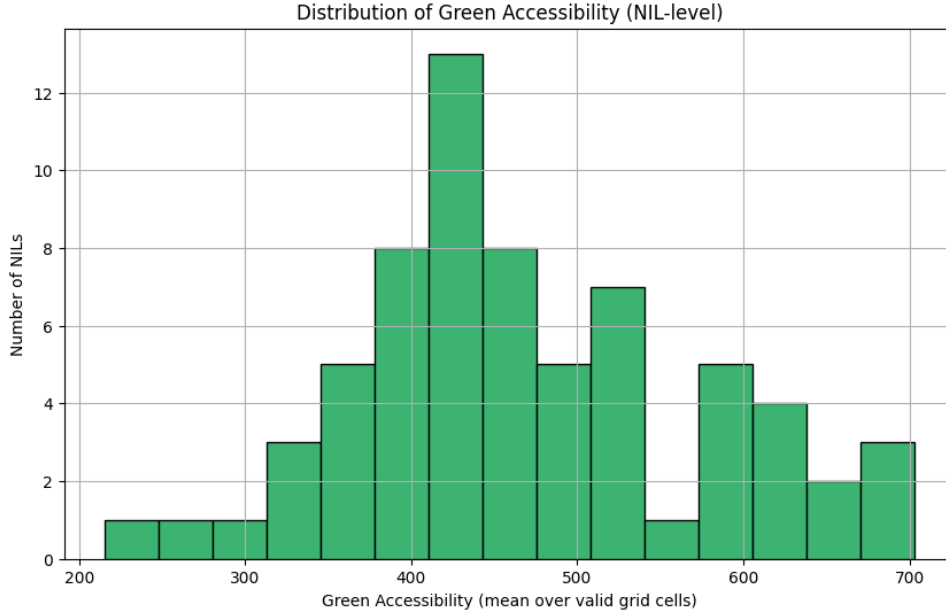


Figure 1: Distribution of Green Accessibility Index (GA) across Milan’s NILs. Each value represents the mean GA computed from valid grid cells within a NIL. The histogram shows notable inter-neighborhood variability and a slight right-skewness.

Spatially, higher accessibility concentrates in the northwestern and northern sectors, supported by large parks such as Parco Nord, Parco di Trenno, and Monte Stella (Figure 2). Many semicentral NILs form a continuous belt of comparatively low accessibility, reflecting dense built fabric and scarce large-scale green infrastructure. Peripheral areas adjacent to linear corridors, such as the Lambro river system, show moderate levels.

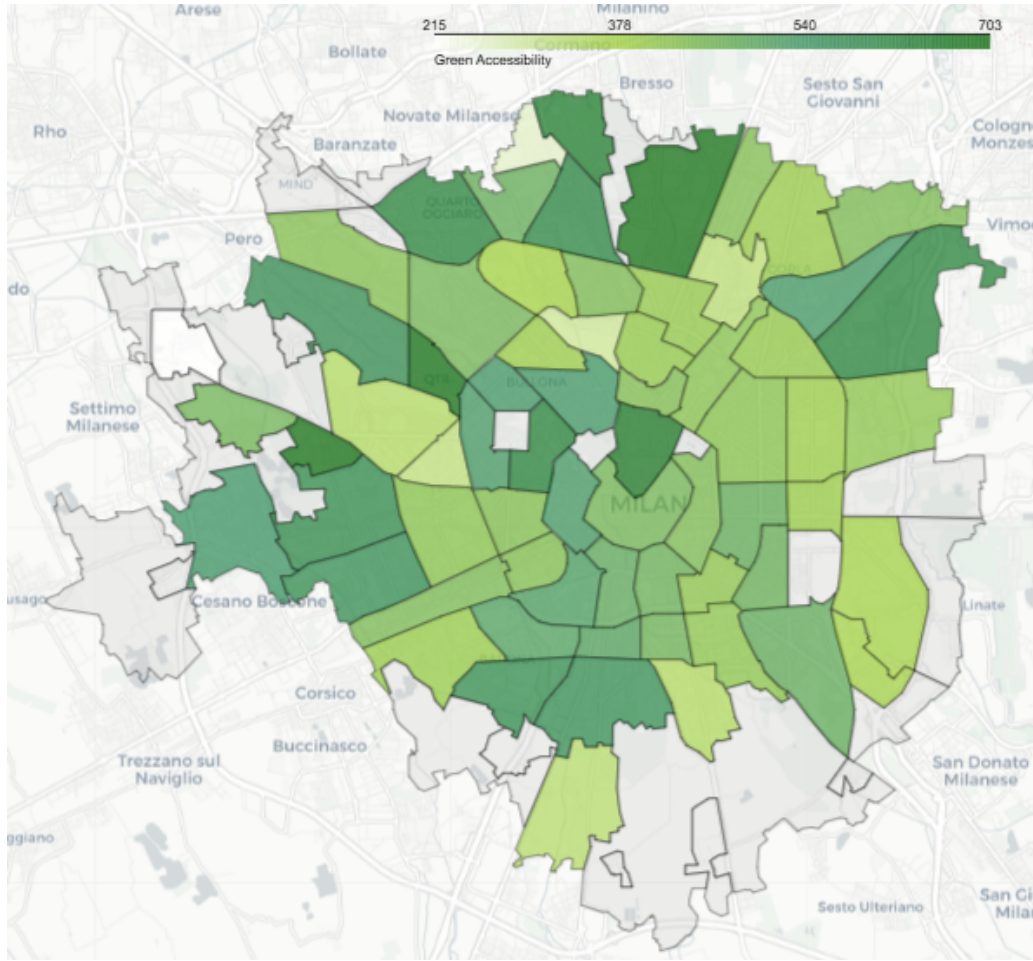


Figure 2: Spatial distribution of Green Accessibility across Milan's NILs. Darker shades indicate higher walkable access to public green space. Grey NILs are excluded according to the criteria in Section 3.4.2.

4.1.2 Population Intensity: demographic load and variability

Population Intensity exhibits a broad spread (Figure 3), with dense residential concentrations in central and semicentral NILs and markedly lower values in peripheral zones. This uneven distribution shapes the demand side of green equity: two areas with similar accessibility may face very different competitive pressures depending on their population load.

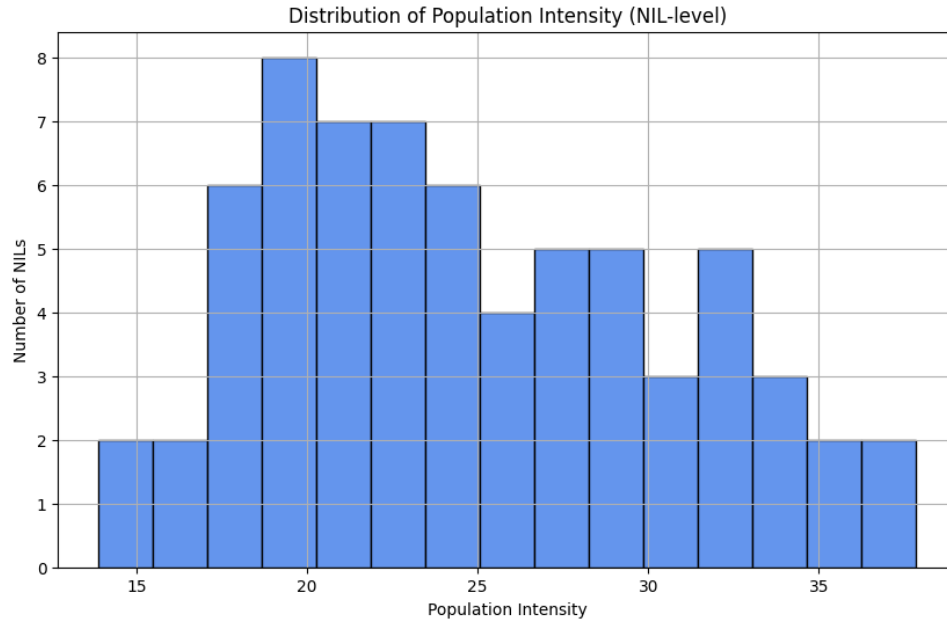


Figure 3: Distribution of Population Intensity across Milan's NILs. Values are square root transformed and represent the average number of residents per grid cell, based on census redistribution.

The spatial pattern (Figure 4) reveals a clear centralization of residents, with peaks in NILs such as Loreto–Casoretto, Porta Venezia, and Porta Magenta. Lower intensities occur in suburban districts with larger parcels or transitional land uses, including parts of QT8, Quinto Romano, and Bicocca.

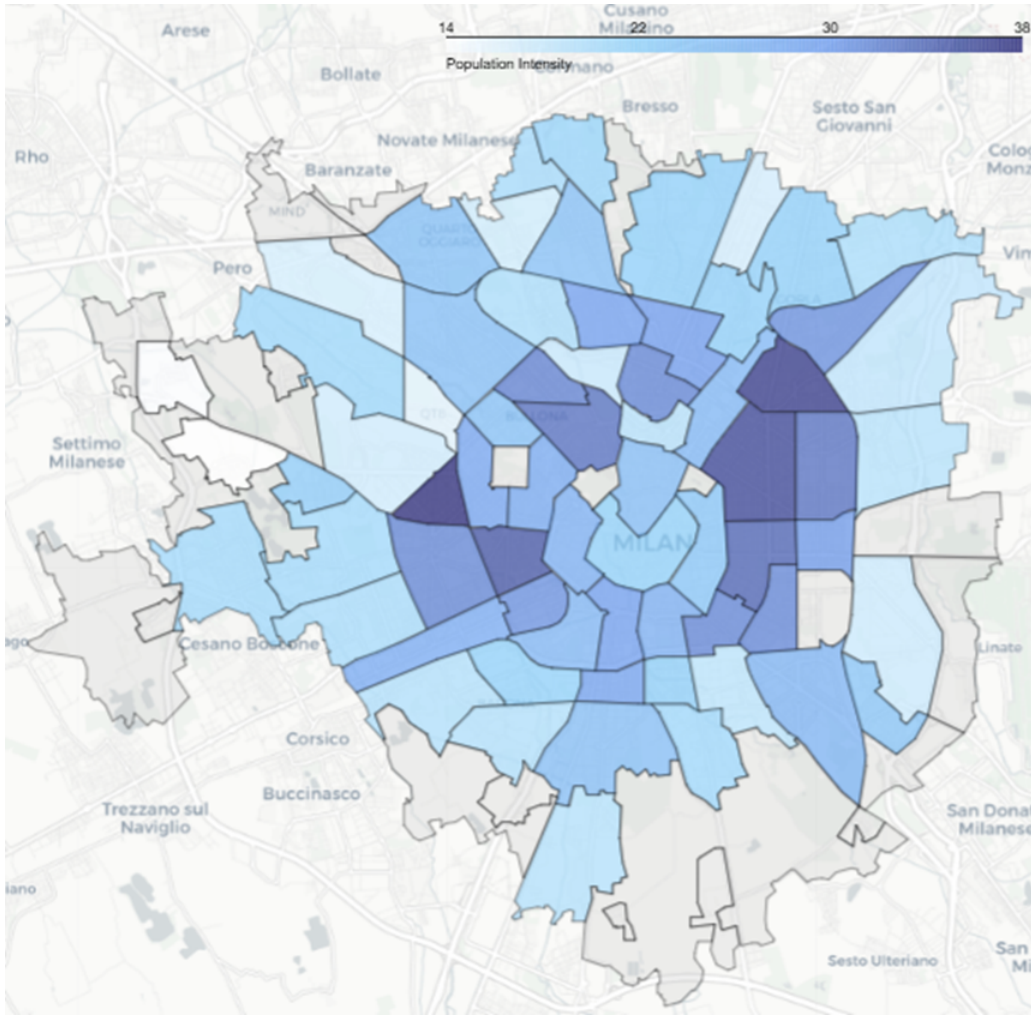


Figure 4: Spatial distribution of Population Intensity across Milan's NILs. Darker shades correspond to higher demographic concentration. Grey NILs are excluded according to the criteria in Section 3.4.2.

4.1.3 Green Equity Ratio (GER): relative balance of supply and demand

GER values are highly variable and strongly right-skewed (Figure 5). High ratios typically correspond to areas combining moderate or high accessibility with low demographic pressure, whereas low ratios emerge where dense populations compete for limited green space.

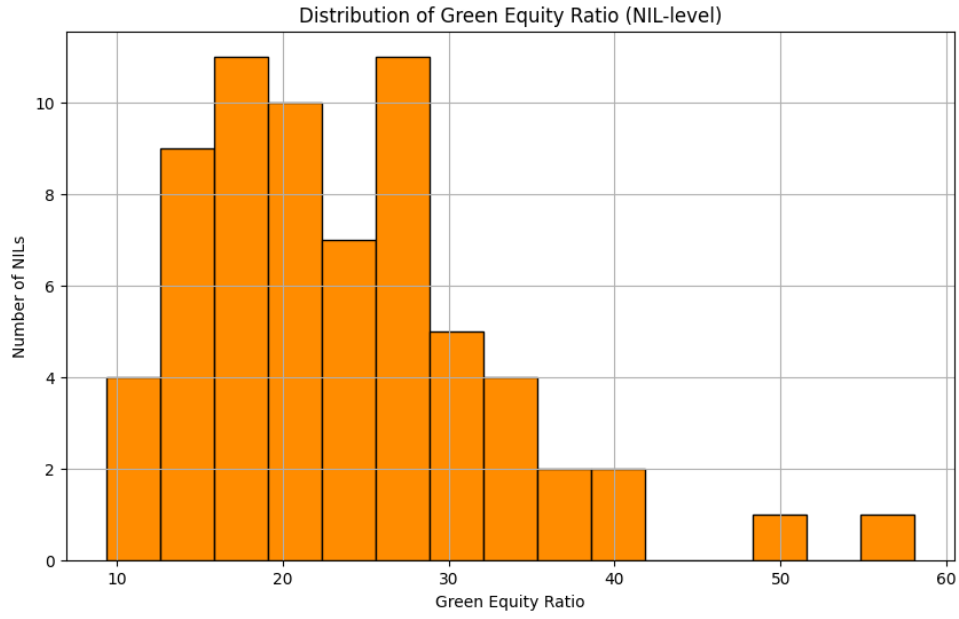


Figure 5: Distribution of Green Equity Ratio (GER) across Milan’s NILs. Each value is the mean GER computed from valid grid cells within a NIL.

The spatial distribution (Figure 6) underscores this contrast: peripheral NILs such as QT8 and Niguarda enjoy favourable balances, while central and semicentral NILs consistently record the lowest ratios. GER thus reveals disparities not evident from accessibility or population intensity alone.

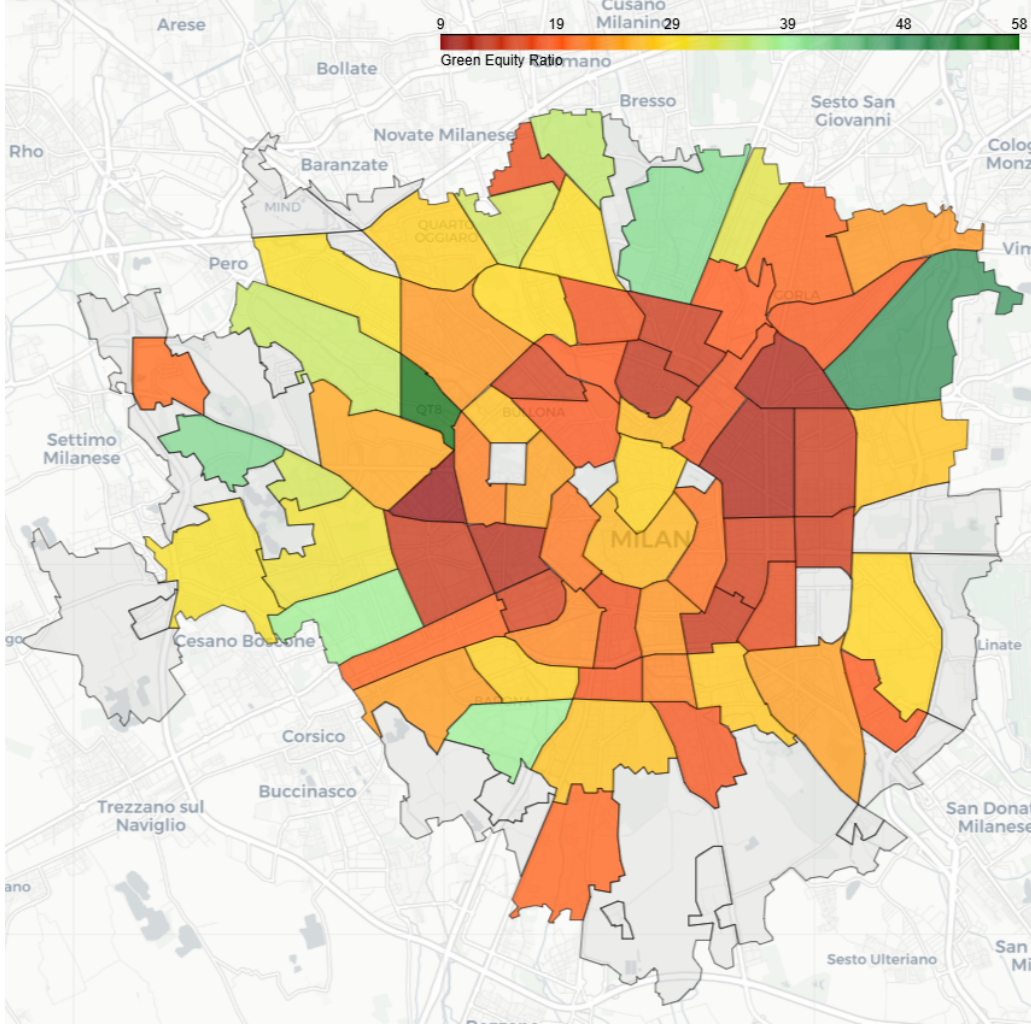


Figure 6: Spatial distribution of GER across Milan’s NILs. Higher values indicate a more favourable balance between green access and local demographic pressure. Grey NILs are excluded according to the criteria in Section 3.4.2.

4.1.4 Relations between measures

The correlation matrix (Table 1) confirms that GER is positively related to Green Accessibility and negatively to Population Intensity, reflecting its role as a relational indicator. By contrast, the near-zero correlation between Accessibility and Intensity shows that supply and demand are spatially independent in Milan.

Table 1: Pairwise Pearson correlations between the three core indicators. Values are expressed as percentages ($\times 100$) and rounded to one decimal place.

	Green Accessibility	Population Intensity	Green Equity Ratio
Green Accessibility	100.0	-3.6	62.8
Population Intensity	-3.6	100.0	-70.7
Green Equity Ratio	62.8	-70.7	100.0

These results validate the decision to use Accessibility and Intensity as orthogonal inputs for clustering, while excluding GER to avoid redundancy. More broadly, they illustrate the importance of a multidimensional approach: analysing supply and demand separately, then jointly, captures inequities that a single metric could obscure.

4.2 NIL Typologies

The typological classification of Milan’s NILs captures recurring patterns in the balance between walkable green accessibility and demographic pressure. Using unsupervised clustering on standardized NIL-level indicators, the analysis identifies latent urban profiles that extend beyond administrative boundaries and reflect data-driven similarities in local accessibility–demand conditions. This typology provides a comparative lens for assessing spatial equity and supports the design of differentiated, evidence-based planning strategies.

Each cluster represents a distinct combination of supply and demand factors, enabling planners to pinpoint areas with acute deficits, structural surpluses, or balanced conditions. The resulting framework strengthens the operational link between quantitative analysis and targeted policy design, anticipating the strategic recommendations discussed in Section 5.

This section presents the clustering results in three parts: (i) evaluation of model performance and parameter selection; (ii) interpretation of the identified NIL types; and (iii) treatment of ambiguous cases with uncertain cluster membership.

4.2.1 Evaluation of Clustering Performance

Clustering was applied to 67 NILs out of the 88 total units in Milan. As outlined in Section 3.4.2, inclusion required at least 10 valid grid cells and a minimum spatial coverage of 40%. This filtering excluded units with insufficient data density, ensuring that cluster assignments were based on statistically robust indicators.

The number of clusters was determined by combining two standard diagnostics: the elbow method, which examines within-cluster inertia, and the average silhouette score, which evaluates cohesion and separation. The elbow plot (Figure 7) shows a clear inflection at $k = 3$, beyond which additional clusters yield diminishing improvements in compactness. Silhouette analysis (Figure 8) also peaks at $k = 3$, indicating the highest overall cluster quality at that value.

Despite these signals, the final model adopts $k = 4$. This choice slightly reduces the global silhouette score (from the peak to 0.414) but introduces a typologically meaningful fourth group—NILs with intermediate profiles that would otherwise be merged into more extreme categories. The trade-off prioritizes interpretive granularity over a marginal loss in statistical compactness, consistent with the framework’s policy-oriented objectives.

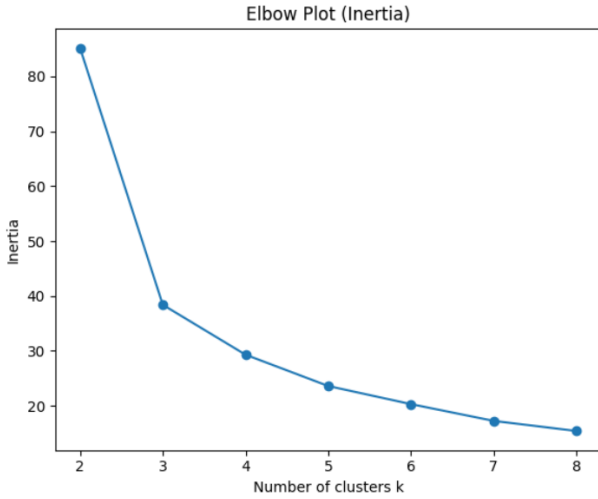


Figure 7: Elbow plot for K-Means clustering. The inflection at $k = 3$ marks the point beyond which compactness gains diminish.

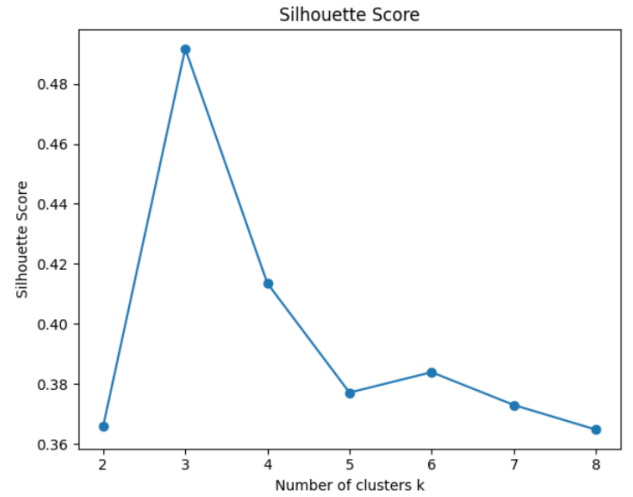


Figure 8: Average silhouette score for different values of k . The peak at $k = 3$ indicates the highest overall cohesion and separation.

The final $k = 4$ solution achieves good overall performance, with all clusters showing average silhouette scores above 0.34 (Table 2). The table reports the mean standardized values of Green Accessibility and Population Intensity—the two clustering inputs—alongside the Green Equity Ratio, which is retained here solely for interpretive purposes in Section 4.2.2. This configuration provides both statistical validity and typological differentiation, ensuring that the resulting NIL profiles capture meaningful variations in accessibility–demand conditions.

Table 2: Cluster-level summary statistics. Green Accessibility (GA) and Population Intensity (PI) are standardized (z) inputs to clustering. Green Equity Ratio (GER) is shown for interpretation only. Silh. Score is the average silhouette coefficient per cluster.

Cluster ID	GA (z)	PI (z)	GER (z)	NIL Count	Silh. Score
0	-0.87	-0.98	0.11	20	0.44
1	-0.46	1.37	-1.06	16	0.43
2	1.47	-0.45	1.21	15	0.43
3	0.16	0.27	-0.21	16	0.34

4.2.2 Interpretation of cluster-based NIL typologies

The NIL typologies are derived from the relative positioning of cluster centroids in a two-dimensional space defined by standardized Green Accessibility (GA) and Population Intensity (PI). GA measures the supply of walkable public green areas, while PI reflects demographic demand. The Green Equity Ratio (GER), excluded from the clustering process, is here used as a complementary indicator to assess supply–demand balance.

The four clusters (Table 2) correspond to distinct urban profiles:

- **Cluster 0 – Low-Pressure, Low-Access:** Low GA and low PI, producing GER values near the citywide mean. Represents areas with minimal provision and minimal demand, typically peripheral or marginal locations.
- **Cluster 1 – Underserved High-Demand:** High PI and below-average GA, resulting in strongly negative GER. Concentrated in dense semicentral and eastern neighborhoods, these areas combine high demand with insufficient access.
- **Cluster 2 – Well-Served Low-Density:** High GA and low PI, yielding a strongly positive GER. Found mainly in peripheral sectors with major parks and low residential pressure.
- **Cluster 3 – Balanced Intermediate:** GA and PI close to the city average, with slightly negative GER. Includes mixed contexts without strong equity imbalances.

The distribution is relatively even: Cluster 0 (30%), Cluster 1 (24%), Cluster 2 (22%), Cluster 3 (24%). This balance supports a diversified typological framework in which no single condition dominates. From a planning perspective, Cluster 1 represents a clear equity deficit and a primary target for corrective action, while Cluster 2 signals a surplus that may raise efficiency considerations. Clusters 0 and 3 indicate intermediate or context-dependent conditions, suggesting that local-scale assessments are necessary before intervention. These distinctions provide a direct operational link to the policy strategies developed in Section 5.

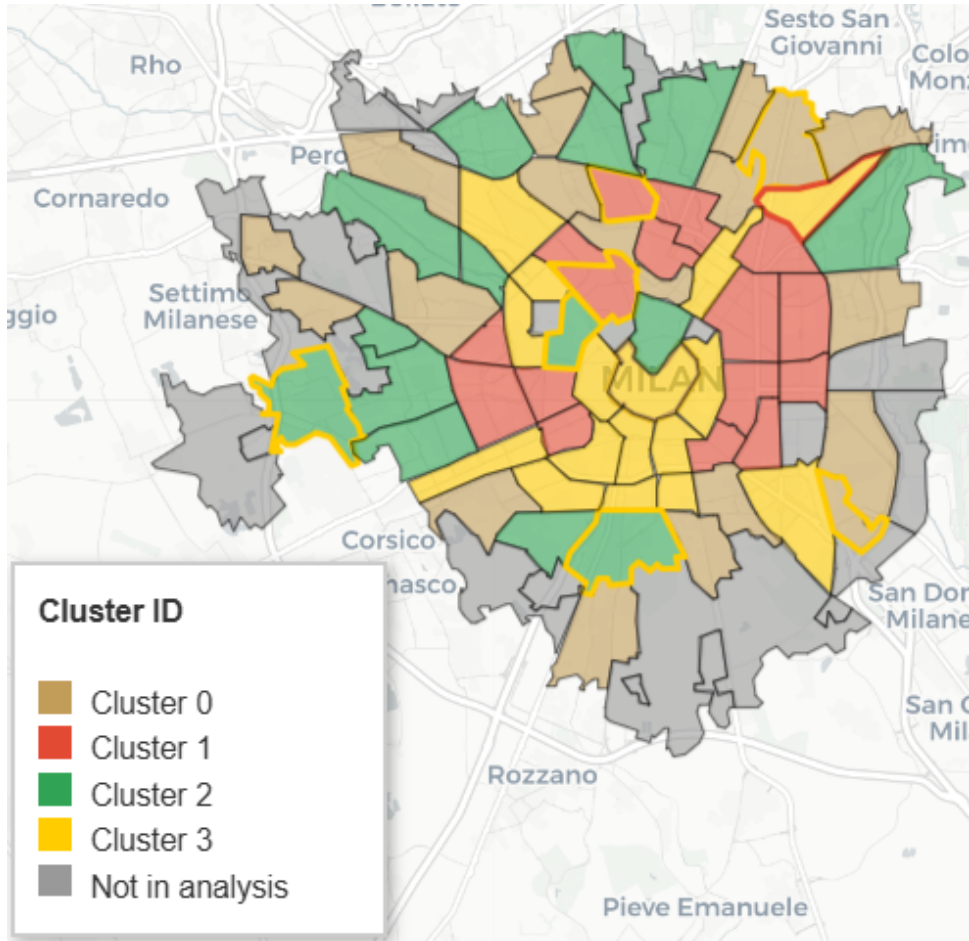


Figure 9: Spatial distribution of NIL typologies across Milan, based on standardized values of GA and PI. Grey areas correspond to NILs excluded from analysis. Ambiguous assignments are indicated by an outer border in the color of their nearest alternative cluster (see Section 4.2.3).

4.2.3 Ambiguity in NIL Assignment

NILs with a silhouette score below 0.20 were flagged as ambiguous and assigned a secondary cluster according to proximity to the next-closest centroid. Out of 67 clustered units, 9 (13.4%) met this criterion, spanning all four typologies. Cluster 2 contains the largest share of ambiguous NILs (3), while Clusters 0, 1, and 3 each contain 2. All ambiguous cases involve transitions with Cluster 3, underscoring its role as a transitional type in the GA-PI space.

In the cluster map (Figure 9), ambiguous NILs are outlined using the color of their secondary cluster, enabling rapid identification of boundary conditions. This representation increases analytical transparency and highlights neighborhoods where accessibility-demand profiles are not strongly polarized. In policy terms, these ambiguous areas may benefit from adaptive or gradient-based interventions rather than rigidly type-specific measures.

5 Policy Implications

This section translates the analytical results into actionable guidance for urban policy and planning. It is structured around three components: (i) overarching design principles for promoting spatial justice in the provision of walkable green areas, (ii) differentiated intervention strategies tailored to the NIL typologies presented in Section 4, and (iii) a critical reflection on the methodological limits and broader social implications of the approach. Throughout, the aim is to align quantitative findings with context-sensitive governance tools, ensuring that policies address both structural inequities and locally specific conditions.

5.1 Policy Design Principles

The framework adopts a dual-axis perspective on spatial justice, grounded in the joint analysis of Green Accessibility (GA) and Population Intensity (PI). GA measures the supply of walkable public green infrastructure, while PI functions as a proxy for local demand. Equity, in this view, emerges not from uniform provision but from proportional adequacy — the balance between supply and demand in each neighborhood [47]. This principle shifts the planning focus from citywide averages to the relational adequacy of provision at the NIL scale.

The three indicators developed in this study — GA, PI, and the Green Equity Ratio (GER) — are not merely diagnostic tools but operational levers. GA quantifies access to green areas within a walkable distance; PI captures the demographic pressure exerted on those resources; GER synthesizes these dimensions into a single measure of relative surplus or deficit. Used together, they enable planners to identify areas of systemic disadvantage (low GA, high PI, negative GER) or underutilized surplus (high GA, low PI, positive GER) and to calibrate interventions accordingly. The use of continuous, standardized indicators supports comparability across neighborhoods and adaptability to evolving urban dynamics.

Beyond individual metrics, the clustering analysis introduced in Section 4 offers a typological lens. By grouping NILs with similar accessibility–demand profiles, it allows planning to move from reactive, case-by-case decisions toward proactive, pattern-based strategies. This typological framing enables resource allocation and policy instruments to be tailored to structural similarities rather than administrative boundaries alone — a practice increasingly adopted in urban equity studies [43].

Finally, the presence of ambiguous NILs (Section 4.2.3) reinforces the importance of flexible instruments. Not all neighborhoods fit neatly into a single type; some exhibit transitional profiles that demand adaptive, hybrid approaches. Policy responses in such cases should incorporate phased interventions, ongoing monitoring, and iterative adjustment rather than rigid, type-specific prescriptions.

5.2 Cluster-Based Interventions

Intervention priorities vary across the four NIL typologies identified in Table 2, each representing a recurring configuration of GA and PI:

- **Cluster 1 — Underserved High-Demand** ($GA_z = -0.46$, $PI_z = 1.37$): These NILs face the most acute equity gap, with high demographic pressure and below-average accessibility, resulting in strongly negative GER values. Strategic priorities include expanding walkable access through micro-parks, adaptive reuse of vacant parcels, targeted upgrades to existing spaces, and improved pedestrian connectivity. These areas require high-priority investment and systematic monitoring of outcomes.
- **Cluster 2 — Well-Served Low-Density** ($GA_z = 1.47$, $PI_z = -0.45$): While equity conditions are favorable, the imbalance between high provision and low demand raises efficiency concerns. The focus here is on preserving assets, optimizing usage, and extending benefits to adjacent, less-served NILs via inter-NIL connections. Interventions are preventative rather than corrective.
- **Cluster 0 — Low-Pressure, Low-Access** ($GA_z = -0.87$, $PI_z = -0.98$): Demand and provision are both low, often in peripheral or infrastructurally marginal locations. Incremental, cost-effective greening (e.g., community-led projects, temporary green uses) can improve livability without overcommitting resources.
- **Cluster 3 — Balanced Intermediate** ($GA_z = 0.16$, $PI_z = 0.27$): These NILs approximate citywide averages, with slightly negative GER. The priority is to preserve the current equilibrium through regular maintenance, targeted improvements, and vigilance for early signs of imbalance. Given their frequent overlap with ambiguous NILs, flexible and data-informed governance is critical.

Across all clusters, three cross-cutting principles enhance implementation: (i) *multi-level coordination* between strategic plans, municipal regulations, and local initiatives; (ii) *timing adaptation*, combining rapid tactical interventions with longer-term investments; and (iii) *flexibility for ambiguity*, using secondary cluster assignments to guide hybrid strategies in transitional contexts.

5.3 Data Limitations and Social Implications

The analytical framework is constrained by both data availability and methodological assumptions. At the grid-cell level, only cells with at least 25 residents were retained (Section 3.4.1). At the NIL level, inclusion required at least 10 valid cells covering a minimum of 40% of the unit’s area (Section 3.4.2). These thresholds safeguard

indicator reliability but also bias coverage toward compact, well-mapped areas, potentially underrepresenting peripheral or irregular zones.

A specific technical gap involves the pedestrian network: in several large parks, unmapped internal paths prevent isochrone-based models from recognizing these spaces as walkable destinations, leading to systematic underestimation of GA. This is most pronounced in low-density sectors with fragmented pedestrian infrastructure.

Indicator-level limitations also affect interpretation. GA treats all green areas equally, without accounting for quality, amenities, or safety; PI relies on static census data, overlooking transient or seasonal populations; GER is sensitive to extreme values and omits subjective perceptions of fairness or accessibility. While these metrics provide a robust quantitative baseline, they cannot substitute for qualitative dimensions of urban experience.

Ethically, the methodology risks excluding populations that are both underserved and underrepresented in datasets — e.g., informal settlements, seasonal workers, unhoused residents. Relying solely on quantifiable indicators may inadvertently reinforce inequities by directing attention and resources toward areas that are well-represented in data rather than those in greatest need.

To mitigate these risks, three complementary strategies are recommended: (i) *participatory methods* such as community mapping and localized surveys to capture unmeasured needs; (ii) *integration of unconventional data sources* (e.g., citizen science platforms, collaborative mapping) to enrich the analytical base; (iii) *adaptive modelling*, treating results as iterative inputs to deliberation rather than fixed outputs. In this perspective, quantitative models become tools for negotiation between evidence and lived experience, ensuring that spatial equity is pursued not just statistically but socially.

6 Conclusions

This study presented a spatially explicit, data-driven framework for assessing walkable green equity in Milan, integrating network-based accessibility, population distribution, and a composite Green Equity Ratio (GER). The approach revealed marked intra-urban disparities: central and semicentral NILs often combine high population intensity with low accessibility, while many peripheral areas show a relative surplus of green provision under lower demographic pressure. The clustering of NILs into typologies provided a structured basis for differentiated interventions, moving beyond uniform standards toward need-based planning.

Methodologically, the framework is modular, transferable, and applicable to other public services where supply-demand balance is critical. Built on open data, standardized spatial units, and reproducible geospatial methods, it supports evidence-based decision-making at multiple scales. The same logic can inform equity assessments for healthcare, education, or mobility infrastructure.

Future work should integrate qualitative and temporal dimensions of access—such as perceived quality, safety, seasonal patterns, or daily usage—to complement quantitative indicators and address experiential aspects of urban equity. Embedding this framework within participatory governance processes can enhance its legitimacy and ensure that spatial justice is pursued in both statistical and social terms.

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